

Variables, Data Types and Input/Output in C++

> **Definition to Remember**

1. Concept Overview & Fundamentals

Variables, Data Types and Input/Output in C++ provides modern high-performance capabilities combining low-level hardware control with high-level abstractions, object-oriented paradigms, and Standard Template Library (STL) utilities.

In modern C++ (C++11 through C++20):

2. Technical Architecture & Memory Mechanics

C++ Runtime Execution Architecture

```text

### Modern C++ Feature Comparison

| Modern C++ Paradigm | Legacy Equivalent | Engineering Advantage | Performance Impact |

---

### 3. Core Syntax & Implementation Patterns

#### Modern C++ Template Structure

```
```cpp
// Fast I/O and standard namespace configuration

std::cout << "Executing C++ Module: Variables, Data Types & InputOutput" << "\n";

return 0;
```
```

---

### 4. Real-World Applications & System Engineering

- **Quantitative Finance & HFT:** High-frequency trading platforms use modern C++ for microsecond-level execution.

---

### 5. Comprehensive Solved Problems & Code Walkthroughs

#### Problem 1: AreaOfCircle

```
Problem Description & Objective:

C++ Code Implementation:
```cpp
// Implementation details...
```

— — —

```
int mai
```

```
cout <<
```

```
cin >>
```

const c

area =

```
cout <<
```

return

Problem 2: AreaOfSquare

****Problem Description & Objective:****

****C++ Code Implementation:****

/*

```
#include <iostream> // Header for standard console I/O stream operations
```

```
using namespace std; // Use standard namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Prompt user to enter side length of the square
```

```
// Read side length from input stream into variable 'side'
```

```
// Calculate the area using formula: Area = side * side
```

```
// Output the resulting area to console
```

```
// Return 0 indicating program finished cleanly
```

return

Problem 3: Circumference

****Problem Description & Objective:****

Implem

****C++ Code Implementation:****

```cpp

/\*

- ===

```
#include <iostream> // Include I/O stream header for console interaction
```

```
using namespace std; // Standard namespace declaration
```

```
// --- Main Execution Entry Point ---
```

int mai

```
// Declare floating-point variables for radius and output circumference
```

double

```
// Output prompt message to console
```

cout &lt;

```
// Read radius value from standard input
```

```
cin >>
```

```
// Apply circle circumference formula: $C = 2 * \pi * r$
```

```
circum
```

```
// Output the final circumference value
```

```
cout <<
```

```
// Return status code 0 to operating system
```

```
return
```

```

```

#### **Problem 4: Declare2number**

**\*\*Problem Description & Objective:\*\***

Implem

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream> // Header for stream-based console input and output
```

```
using namespace std; // Standard library namespace
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

```
// Declare integer variable 'num2' and assign initial value 25
```

```
int num
```

```
// Print the value stored in num1
```

```
cout <
```

```
// Print the value stored in num2
```

```
cout <
```

```
// End main function with return code 0
```

```
return
```

```
---
```

Problem 5: NameAge

****Problem Description & Objective:****

Implem

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream> // Header for stream I/O operations
```

```
#includ
```

```
using namespace std; // Standard namespace declaration
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

// Declare integer variable for age

int age

// Prompt and read first name

cout <

// Prompt and read last name

cout <

// Prompt and read integer age

cout <

// Display formatted user information by combining string streams

cout <

// Program execution finished successfully

return

---

### **Problem 6: PatternPrint1**

**\*\*Problem Description & Objective:\*\***

Implem

**\*\*C++ Code Implementation:\*\***

```cpp

/*

- ===

```
#include <iostream> // Console stream header

using namespace std; // Use std namespace

// --- Main Execution Entry Point ---
```

```
int main
```

```
// Return zero status code
```

```
return
```

```
---
```

Problem 7: PatternPrint2

****Problem Description & Objective:****

Implemen

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream> // Header for console stream output

using namespace std; // Standard namespace

// --- Main Execution Entry Point ---
```

```
// Successful return
```

```

```

### Problem 8: SizeOf

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream> // Header for console I/O

using namespace std; // Standard namespace

// --- Main Execution Entry Point ---
```

```
int main
```

```
// Evaluate memory size of integer variable in bytes
```

```
cout <
```

```
// Evaluate memory size of float variable in bytes
```

```
cout <
```

```
// Evaluate memory size of double variable in bytes
```

```
cout <
```

```
// Evaluate memory size of char variable in bytes
```

```
cout <
```

```
// Evaluate memory size of bool variable in bytes
```

```
cout <
```

```
// Return zero indicating normal finish
```

```
return
```

```

```

### **Problem 9: Swap**

**\*\*Problem Description & Objective:\*\***

Implem

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream> // Header for input/output operations
```

```
using namespace std; // Standard namespace
```

```
// --- Main Execution Entry Point ---
```

```
int main
```

```
// Read first number 'a'
```

```
cout <<
```

```
// Read second number 'b'
```

```
cout <<
```

```
// Display values before swap operation
```

```
cout <<
```

```
// --- SWAPPING MECHANISM ---
```

```
// Step
```

```
// Step 2: Assign value of 'b' into 'a' (overwriting original 'a')
```

```
a = b;
```

```
// Step 3: Assign saved original value from 'temp' into 'b'
```

```
b = temp
```

```
// Display values after swap operation
```

```
cout << endl
```

```
// Program completed successfully
```

```
return 0
```

```
---
```

Problem 10: Welcome

****Problem Description & Objective:****

Implement

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream> // Console stream header
```

```
#include
```

```
using namespace std; // Standard namespace
```

```
// --- Main Execution Entry Point ---
```

```
int main
```

```
// Output prompt for user name
```

```
cout <<
```

```
// Use std::getline to capture full name line including spaces
```

```
getline
```

```
// Print welcome message incorporating captured name
```

```
cout <<
```

```
// Return zero code
```

```
return
```

```

```

## 6. Exam Scoring Points & Memory Revision

```
> **Must-Write Points**
```

```
> 1. Ty
```

```
> **Quick Recall**
```

```
> `std:
```